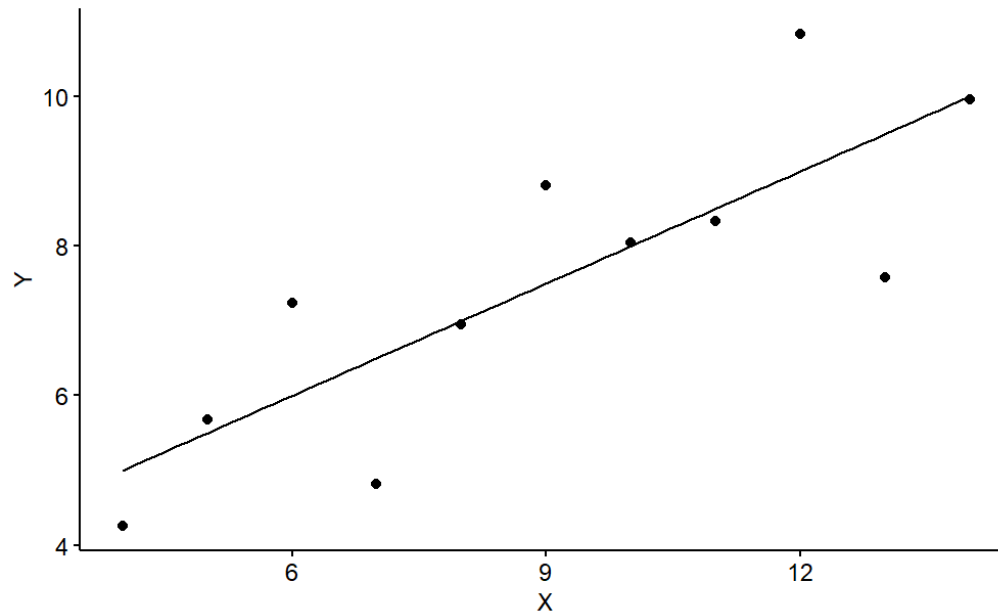


A data-based illustration of leverage, discrepancy, and influence

To understand the concepts of leverage, discrepancy, and influence further, let's consider a made-up dataset of only $N = 11$ cases; these cases comprise “dataset 1” from Anscombe's (1973) famous fictitious datasets. In this dataset, there is a clear positive relation between X and Y which is well-represented by a simple linear regression model, as depicted in Figure S1.1:

Figure S1.1



The OLS estimated regression equation from these data is $\hat{Y}_i = 3.00 + 0.50X_i$, with $R^2 = .67$. The data, along with the Studentized residuals, hat values, and Cook's D resulting from this model are in Table S1.1. At this point, we simply note that none of the Studentized residuals or hat values seems particularly extreme, and consequently neither are the Cook's D s, which is not surprising given the apparent lack of outlying observations in Figure S1.1.

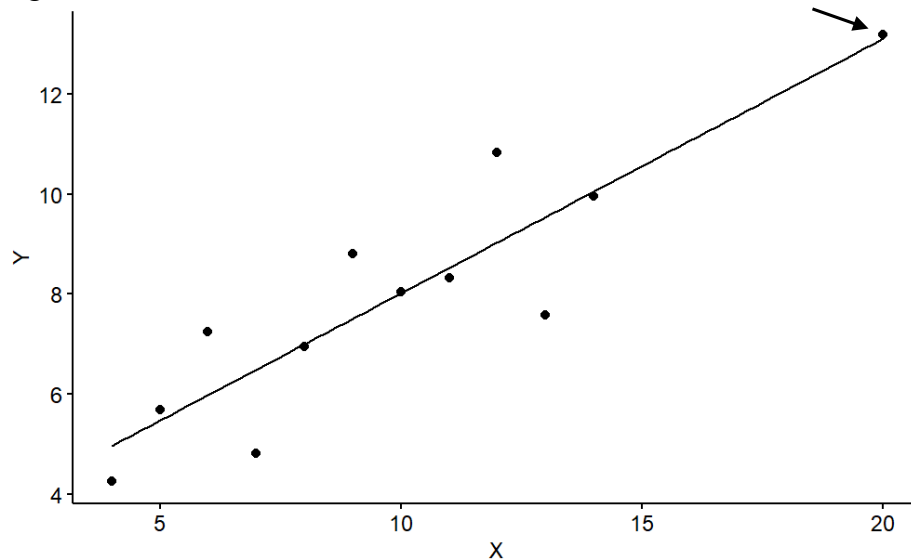
Table S1.1. Data for original $N = 11$ cases

Case i	X	Y	e_i^*	h_i	D_i
1	10	8.04	0.03324	0.00909	0.00006
2	8	6.95	-0.04332	0.00909	0.0001
3	13	7.58	-1.77793	0.14545	0.48921
4	9	8.81	1.11029	0	0.06164
5	11	8.33	-0.1481	0.03636	0.0016
6	14	9.96	-0.04051	0.22727	0.00038
7	6	7.24	1.1019	0.08182	0.12676
8	4	4.26	-0.72516	0.22727	0.1227
9	12	10.84	1.63487	0.08182	0.27903
10	7	4.82	-1.45488	0.03636	0.15434
11	5	5.68	0.16607	0.14545	0.00427

Note. e^* is Studentized residual; h is hat value; D is Cook's distance.

Next, I added a 12th case (case $i = 12$) that has a large leverage but a small discrepancy, or residual, from the estimated regression model. This case appears as a clear outlier in Figure S1.2 (the arrow is pointing at the outlier):

Figure S1.2



With this outlying case added to the data, the estimated regression equation becomes $\hat{Y}_i = 2.93 + 0.52X_i$, which is very similar to the original equation, but now $R^2 = .81$. The data with the new case along with the new Studentized residuals, hat values, and Cook's D s are in Table S1.2. Here, we see that case 12 has a much larger leverage ($h_{12} = .46$) than the other cases. But because its residual is small ($e_{12}^* = .11$), it has very little influence ($D_{12} = .01$). Consequently, the estimated regression equation changes very little when this case is added to the data. But this case has also increased R^2 from .67 to .81 and decreased the standard error of the regression slope from 0.12 to 0.08. For this reason, a high-leverage, low-discrepancy case is often referred to as a “good leverage” case because it improves the *precision* of the parameter estimates (“precision” refers to sampling variability of the estimates, as measured by their standard error).

Table S1.2

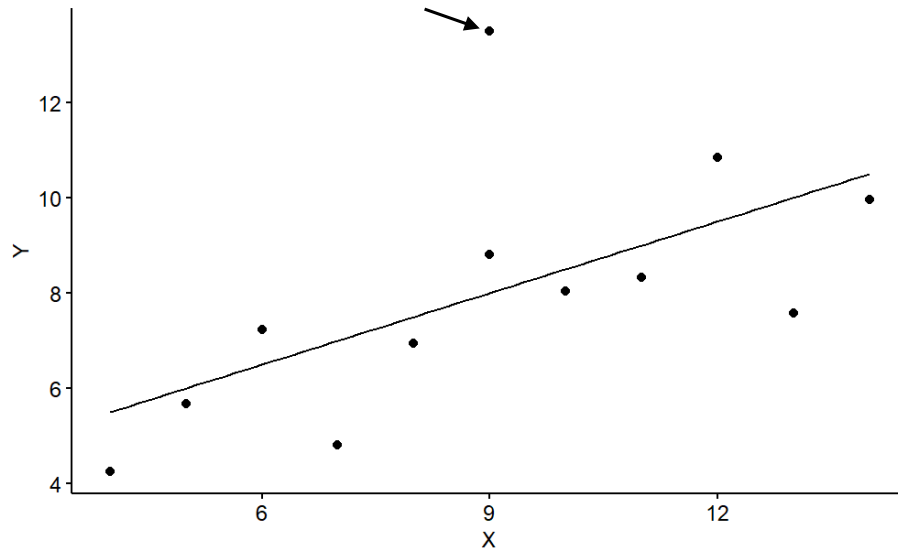
Data for original $N = 11$ cases plus high leverage, low discrepancy case ($i = 12$)

Case i	X	Y	e_i^*	h_i	D_i
1	10	8.04	.01934	.00003	.00002
2	8	6.95	-.04489	.01663	.00011
3	13	7.58	-1.79147	.04303	.23211
4	9	8.81	1.15984	.00380	.06420
5	11	8.33	-.17614	.00531	.00151
6	14	9.96	-.08804	.07547	.00073
7	6	7.24	1.16450	.06944	.12226
8	4	4.26	-.68820	.15846	.07552
9	12	10.84	1.62209	.01965	.15103
10	7	4.82	-1.51887	.03851	.16004
11	5	5.68	.19664	.10942	.00462
12	20	13.20	.11400	.46024	.00774

Note. e^* is Studentized residual; h is hat value; D is Cook's distance.

Figure S1.3 shows the same data, except now there is a different outlier; here case 12 has low leverage coupled with high discrepancy:

Figure S1.3



With this outlying case included, the estimated regression equation is $\hat{Y}_i = 3.50 + 0.50X_i$. Although the outlier has shifted the intercept, the slope estimate is identical to the initial model, but $R^2 = .37$. The corresponding Studentized residuals, hat values, and Cook's D s are in Table S1.3. Now, case 12 has a very large Studentized residual ($e_{12}^* = 2.66$), but because it has no leverage ($h_{12} = .00$), it does not have particularly large influence ($D_{12} = .32$). Although this Cook's D is larger than those for the other 11 cases, it is still small in an absolute sense, as Cook's D values are often not considered large until they are near 1.00 or greater. Yet, this case has substantially reduced R^2 from that obtained with the original 11 cases, and the standard error of the slope has increased from .12 to .21. Thus, cases with large discrepancy are harmful for the precision of the parameter estimates even when they have little influence.

Table S1.3

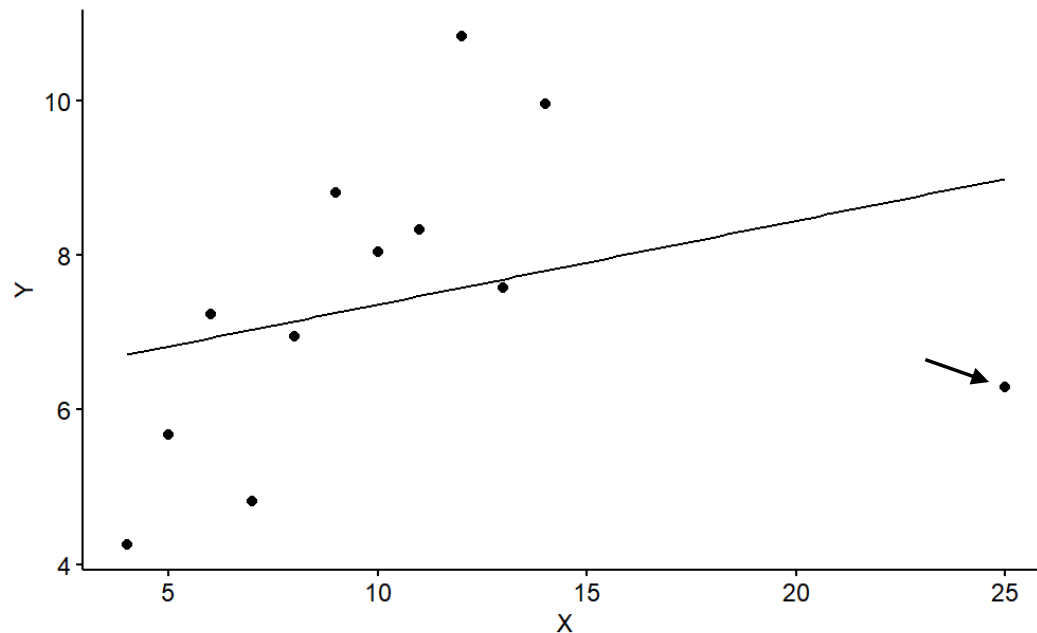
Data for original $N = 11$ cases plus low leverage, high discrepancy case ($i = 12$)

Case i	X	Y	e_i^*	h_i	D_i
1	10	8.04	-.22376	.00909	.00255
2	8	6.95	-.26736	.00909	.00364
3	13	7.58	-1.27509	.14545	.24116
4	9	8.81	.39087	.0	.00694
5	11	8.33	-.33076	.03636	.00744
6	14	9.96	-.30150	.22727	.02048
7	6	7.24	.37428	.08182	.01386
8	4	4.26	-.69090	.22727	.10753
9	12	10.84	.67770	.08182	.04543
10	7	4.82	-1.07490	.03636	.07855
11	5	5.68	-.16877	.14545	.00422
12	9	13.50	2.65637	.0	.32074

Note. e^* is Studentized residual; h is hat value; D is Cook's distance.

Finally, Figure S1.4 shows the data with an outlying case that has both large leverage and large discrepancy, and consequently large influence:

Figure S1.4



Here, as shown in Table S1.4, case 12 clearly has an extreme Studentized residual ($e_{12}^* = -2.54$), large leverage ($h_{12} = .62$), and hence very large influence ($D_{12} = 7.77$). Thus, this outlier has radically affected the estimated regression equation, which is

$$\hat{Y}_i = 6.28 + 0.11X_i,$$

and has reduced R^2 to .10. No doubt, an outlier with large discrepancy and large leverage is a “bad leverage” case!

Table S1.4

Data for original $N = 11$ cases plus high influence case ($i = 12$)

Case i	X	Y	e_i^*	h_i	D_i
1	10	8.04	.35938	.00032	.00590
2	8	6.95	-.10609	.01580	.00062
3	13	7.58	-.05929	.02063	.00020
4	9	8.81	.82906	.00516	.03336
5	11	8.33	.45620	.00129	.00962
6	14	9.96	1.17522	.03901	.09626
7	6	7.24	.16970	.05448	.00230
8	4	4.26	-1.39718	.11638	.24357
9	12	10.84	1.74128	.00806	.15249
10	7	4.82	-1.20204	.03224	.09440
11	5	5.68	-.63704	.08253	.04035
12	25	6.30	-2.53549	.62411	7.77290

Note. e^* is Studentized residual; h is hat value; D is Cook’s distance.

Reference

Anscombe, F.J. (1973). Graphs in statistical analysis. *American Statistician*, 27, 17-21.